The buffer must deliver the data to the user, at the correct bit rate for the media being used. Data which is sent into the buffer should be sent at a higher rate to allow for unexpected delays. The media player continuously monitors how full the buffer is and controls the bit rate in relation to the defined high- and low-water marks. It is essential to have a buffer size that is sufficiently large for it never to get filled.

The rate of transmission to the buffer is limited by the bandwidth of the network connection. For a connection via a PSTN, a broadband link is essential. For good-quality movie presentation the broadband requirement is about 2.5 Mbps. Because this will not be available for all users it is often the practice that an individual video is made available at different levels of compression. The most highly compressed version will be the poorest quality but the bit rate may be sufficiently low for a reasonable presentation with a relatively low bandwidth Internet connection.

TASK 2.02

Consider a bit-streaming scenario for a video where the following values apply:

- the buffer size is 1 MiB
- the low-water mark is set at 100 KiB
- the high-water mark is set at 900 KiB
- the incoming data rate is 1 Mbps
- the video display rate is 300 Kbps.

Assume that the video is playing and that the buffer content has dropped to the low-water mark. The media player sets the controls for data input to begin again.

Calculate the amount of data that will be input to the buffer in two seconds and the amount of data that will be removed from the buffer in the same time period.

Repeat the calculation for 4, 6, 8, 10 and 12 seconds.

From this data, estimate when the buffer will have filled up to the high-water mark.

Assuming that the incoming transmission is halted at this time, calculate how long it will be before the buffer content has again fallen to the low-water mark level.

2.08 IP addressing

The Internet requires technical protocols to function. A protocol suite called TCP/IP is used as a standard (see Chapter 17). One aspect of this is IP addressing, which is used to define from where and to where data is being transmitted.

IPv4 addressing

Currently the Internet uses Internet Protocol version 4 (IPv4) addressing. IPv4 was devised in the late 1970s, before the invention of the PC and the mobile phone. IPv4 provides for a large but limited number of addresses for devices, which is no longer enough to cover all the devices expected to use the Internet in future.

The IPv4 addressing scheme is based on 32 bits (four bytes) being used to define an **IPv4 address**. It is worth putting this into context. The 32 bits allow 2^{32} different addresses. For big numbers like this it is worth remembering that 2^{10} is approximately 1000 in denary so the



KEY TERM

IPv4 address: a 32-bit long, hierarchical address of a device on the Internet

32 bits provide for approximately four billion addresses. The population of the world is about seven billion and it is estimated that approaching half of the world's population has Internet access. From this we can see that if there was a need to supply one IP address per Internet user the scheme would just about be adequate. However, things are not that simple.

The original addressing scheme was designed on the basis of a hierarchical address with a group of bits defining a network (a netID) and another group of bits defining a host on that network (a hostID). The aim was to assign a unique, universally recognised address for each device on the Internet. The separation into two parts allows the initial transmission to be routed according to the netID. The hostID only needs to be examined on arrival at the identified network. Before proceeding, it is important to note that the term 'host' is a little misleading because some devices, particularly routers, have more than one network interface and each interface requires a different IP address.

The other feature of the original scheme was that allocated addresses were based on the concept of different classes of networks. There were five classes; we are going to look at the first three classes. The structures used for the addresses are shown in Table 2.02.

Class	Class identifier	Number of bits for netID	Number of bits for hostID
Class A	0	7	24
Class B	10	14	16
Class C	110	21	8

Table 2.02 Address structure for three classes of IPv4 address

It can be seen from Table 2.02 that the most significant bit or bits identify the class. A group of the next most significant bits define the netID and the remaining, least significant, bits define the hostID. The reasoning behind this was straightforward. The largest organisations would be allocated to Class A. There could only be 2^7 i.e. 128 of these but there could be 2^{24} distinct hosts for each of them. This compared with 2^{21} (approximately two million) organisations that could be allocated to Class C but each of these could only support 2^8 i.e. 256 hosts.

The problems with this scheme arose once LANs supporting PCs became commonplace. The number of Class B netIDs available was insufficient but if organisations were allocated to Class C the number of hostIDs available was too small. There have been a number of different modifications made available to solve this problem.

Before considering some of these, the representation used for an IP address needs to be introduced. During transmission, the technology is based on the 32-bit binary code for the address; to make it simpler for users, we write the address using decimal numbers separated by dots. Each byte is written as the denary equivalent of the binary number represented by the binary code. For example, the 32 bit code:

```
10000000 00001100 00000010 00011110
```

is written in dotted decimal notation as:

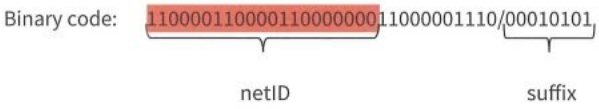
```
128.12.2.30
```

Discussion Point:

There were options available when the dotted decimal notation was chosen. Can you identify these?

Classless inter-domain routing (CIDR)

The first approach developed for improving the addressing scheme is called 'classless inter-domain routing' (CIDR). This retains the concept of a netID and a hostID but removes the rigid structure and allows the split between the netID and the hostID to be varied to suit individual need. The simple method used to achieve this is to add an 8-bit suffix to the address that specifies the number of bits for the netID. If, for instance, we define the suffix as 21, that means that 21 bits are used for the netID and there are 11 bits remaining (of a 32-bit address) to specify hostIDs allowing 2^{11} (i.e. 2048) hosts. One example of an IP address using this scheme is shown in Figure 2.10. The 21 bits representing the netID have been highlighted. The remaining 11 bits represent the hostID which would therefore have the binary value 11000001110.



Dotted decimal notation: 195.12.6.14/21

Figure 2.10 A CIDR IPv4 address

Note that with this scheme there is no longer any need to use the most significant bit or bits to define the class. However, it does allow already existing Class A, B or C addresses to be used with suffixes 8, 16 or 24, respectively.

TASK 2.03

Create an example of the binary code for a Class C address expressed in CIDR format. Give the corresponding dotted decimal representation.

Sub-netting

Sub-netting is a different approach. It allows a more efficient use of a hostID by applying a structure to it.

To illustrate an example of this we can consider a medium-sized organisation with about 150 employees each with their own computer workstation. Let's assume that there are six individual department LANs and one head-office LAN. Figure 2.11 shows a schematic diagram of how the LANs would be connected to the Internet if the original scheme were used. Note that the diagram has been simplified by showing the LANs connected to a gateway. This is a device that connects networks with different protocols. For the connection to the Internet the gateway would either first connect to a router or have the capability to act as a router itself.

The organisation would need seven individual Class C netIDs; one for each LAN. Each of these would point to one of the LAN gateways. The netID for each LAN would be identified by the first 24 bits of the IPv4 address, leaving 8 bits for the hostID. This would mean 256 individual codes for identifying different workstations on just one LAN. For the seven LANs the total number of workstations that could be identified would be:

$$256 \times 7 = 1792$$

Since the organisation only has 150 workstations in total, there are 1642 unused addresses. Not only would these be unused they would be unavailable for use by any other organisation.

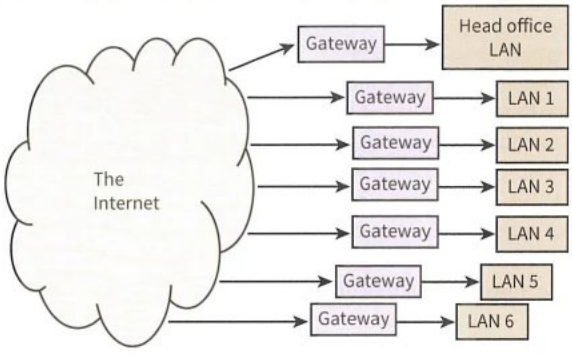


Figure 2.11 Connecting LANs using the original classful IPv4 scheme

The sub-netting solution for this organisation would require allocating just one Class C netID. For example, the IP addresses allocated might be 194.10.9.0 to 194.10.9.255 where the netID comprises the first three bytes, represented by the decimal values 194, 10 and 9.

The sub-netting now works by having a defined structure for the 256 codes constituting the hostID. A sensible solution for this organisation is to use the top three bits as a code for the individual LANs and the remaining five bits as codes for the individual workstations. Figure 2.12 shows a schematic diagram of this arrangement.

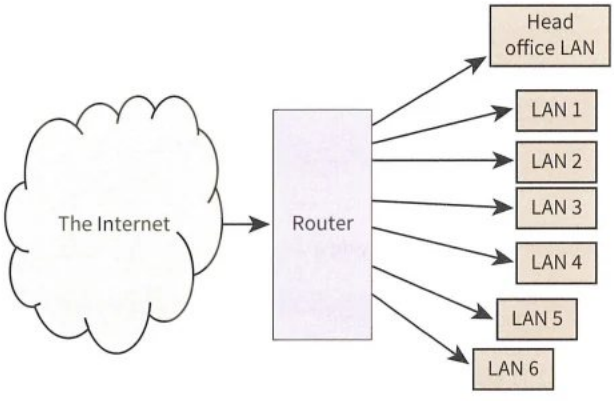


Figure 2.12 Connecting LANs using sub-netting

On the Internet, all of the allocated IP addresses have a netID pointing to the router. The router then has to interpret the hostID to direct the transmission to the appropriate workstations on one of the LANS via a gateway. Examples of workstation identification:

- hostID code `00001110` would be the address for workstation 14 on the head office LAN 0 (LAN 000)
- hostID code `01110000` would be the address for workstation 16 on LAN 3 (LAN 011).

With 150 workstations the organisation hasn't used all of the 256 allocated IP addresses. However, there are only 106 unused which is a reasonable number to have available in case of future expansion. Only one netID has been used leaving the other six that might have been used still available for other organisations to use.

Network address translation (NAT)

The final scheme to be considered is different in that it deviates from the principle that every IP address should be unique. In this scheme, provision has been made for large organisations to have private networks (intranets) which use the same protocols as those used for the Internet. One justification for using a private network has always been that this provides extra security because of the isolation from the Internet. However, this is no longer normal practice. Organisations want private networks but they also want Internet connectivity.

The solution for dealing with the addressing is to use network address translation (NAT). Figure 2.13 shows a schematic diagram of how this can be used.

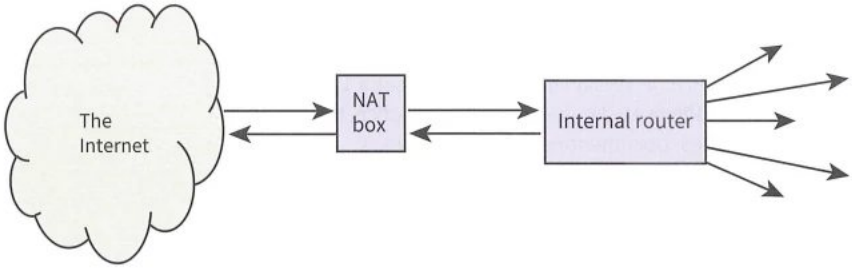


Figure 2.13 An intranet connected to the Internet using a NAT box

The NAT box has one IP address which is visible over the Internet and so can be used as a sending address or as a receiving address. Internally the IP addresses have to be chosen from one of the three ranges of IP addresses shown in Table 2.03 that have been allocated for such networks. (You do not need to remember these numbers!)

Lower bound	Upper bound
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

Table 2.03 IPv4 addresses to be used in private networks

The important point is that each address can be simultaneously used by any number of different private networks. There is no knowledge of such use on the Internet itself or in any other private network. The interface in the NAT box has software installed to examine each incoming or outgoing transmission. There can be a security check before an incoming transmission is directed to the correct internal address. The diagram shows undefined arrows from the router connected to the NAT box. These indicate that the network structure within the organisation could take many different forms.

Static and dynamic IP addresses

As discussed in Section 2.06, when a user wishes to have a connection to the Internet the connection is handled by an Internet Service Provider. The ISP will have available a large number of hostIDs. However, the number of users that the ISP is supporting could very likely be larger than the total number of addresses available. Fortunately for the ISP and for an individual user many of these potential users will not be engaged in Internet interaction.

The normal practice is for the ISP to create a 'dynamic address' for a user. This is one that the ISP is free to change if it suits but more importantly the address is available for re-allocation once a user disconnects from the Internet. The alternative is a 'static address' which never changes and can be provided if a user is prepared to pay an extra charge.

Discussion Point:

Can you find out which IP addressing scheme is being used when you are connected to the Internet?

IPv6 addressing

Today there are combinations of IPv4 approaches in use and these allow the Internet to continue to function. Respected sources argue that this cannot continue beyond the current decade. There must soon be a migration to IP version 6 (IPv6), which uses a 128-bit addressing scheme allowing 2^{128} different addresses, a huge number! In practice, this will allow more complex structuring of addresses. Documenting these addresses is going to be difficult. The addresses are written in a colon hexadecimal notation. The code is broken into 16-bit parts, with each part represented by four hexadecimal characters. Fortunately, some abbreviations are allowed. A few examples are given in Table 2.04.

IPv6 address	Comment
68E6:7C48:FFFE:FFFF:3D20:1180:695A:FF01	A full address
72E6::CFFE:3D20:1180:295A:FF01	:0000:0000: has been replaced by ::
6C48:23:FFFE:FFFF:3D20:1180:95A:FF01	Leading zeros omitted
::192.31.20.46	An IPv4 address used in IPv6

Table 2.04 Some examples of IPv6 addresses

Extension Question 2.01

If IPv6 addressing is used, how many addresses would be available per square metre of the Earth's surface? Do you think there will be enough to go round?

2.09 Domain names

In everyday use of the Internet, a user needs to identify a particular web page or email box. As users, we would much prefer not to identify each IP address using its dotted decimal value! To get round this problem the **domain name service (DNS, also known as domain name system)** was invented in 1983. The DNS service allocates readable domain names for Internet hosts and provides a system for finding the IP address for an individual domain name.



KEY TERM

Domain name service (DNS): a hierarchical distributed database installed on domain name servers that is responsible for mapping a domain name to an IP address. Also known as domain name system.

The system is set up as a hierarchical distributed database which is installed on a large number of domain name servers covering the whole of the Internet. The domain name servers are connected in a hierarchy, with powerful root servers at the top of the hierarchy supporting the whole Internet. The root servers are replicated, meaning that multiple copies of all their data are kept at all times. DNS name space is then divided into non-overlapping zones. Each zone has a primary name server with the database stored on it. Secondary servers get information from this primary server.

As a result, the naming system is hierarchical. There are more than 250 top-level domains which are either generic (e.g. .com, .edu, and .gov) or represent countries (e.g. .uk and .nl).

The domain name is included in a universal resource locator (URL), which identifies a web page, or an email address. A domain is named by the path upward from it. For example, .eng.cisco.com. refers to the .eng subdomain in the .cisco domain of the .com top-level domain.

Looking up a domain name to find an IP address is called 'name resolution'. For such a query there are three possible outcomes.

- If the domain is under the control of the server to which the query is sent then an authoritative and correct IP address is returned.
- If the domain is not under the control of the server, an IP address can still be returned if it is stored in a cache of recently requested addresses but it might be out of date.
- If the domain in the query is remote then the query is sent to a root server which can provide an address for the name server of the appropriate top-level domain. This in turn can provide the address for the name server in the next lower domain. This continues until the query reaches a name server that can provide an authoritative IP address.

Reflection Point:

In several places you have been asked to carry out some research. Are you using the most efficient search methods? Specifically, how could they be improved?

Summary

- Client-server and peer-to-peer networking are options for file sharing.
- The star topology is the one most commonly used for a LAN.
- The main transmission media are copper (twisted pair, coaxial) cables, fibre-optic cables and wireless (radio, microwave, infrared).
- Factors to consider when choosing a medium are bandwidth, attenuation, interference and the need for repeaters.
- CSMA/CD (carrier sense multiple access with collision detection) has been used to detect and avoid message collisions in shared media.
- The Internet is the largest internetwork in existence.
- ISPs provide access to the Internet.
- Internet infrastructure is supported by PSTNs and cell phone companies.
- The World Wide Web is a distributed application accessible on the Internet.
- The current Internet addressing scheme is IPv4, with IPv6 a future contender.
- The DNS resolves a domain name to an IP address.